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Somatic mosaic *IDH1* and *IDH2* mutations are associated with enchondroma and spindle cell hemangioma in Ollier disease and Maffucci syndrome

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Ollier disease and Maffucci syndrome are non-hereditary skeletal disorders characterized by multiple enchondromas (Ollier disease) combined with spindle cell hemangiomas (Maffucci syndrome). We report somatic heterozygous mutations in *IDH1* (c.394C>T encoding an R132C substitution and c.395G>A encoding an R132H substitution) or *IDH2* (c.516G>C encoding R172S) in 87% of enchondromas (benign cartilage tumors) and in 70% of spindle cell hemangiomas (benign vascular lesions). In total, 35 of 43 (81%) subjects with Ollier disease and 10 of 13 (77%) with Maffucci syndrome carried *IDH1* (98%) or *IDH2* (2%) mutations in their tumors. Fourteen of 16 subjects had identical mutations in separate lesions. Immunohistochemistry to detect mutant IDH1 R132H protein suggested intraneoplastic and somatic mosaicism. *IDH1* mutations in cartilage tumors were associated with hypermethylation and downregulated expression of several genes. Mutations were also found in 40% of solitary central cartilaginous tumors and in four chondrosarcoma cell lines, which will enable functional studies to assess the role of *IDH1* and *IDH2* mutations in tumor formation.

Enchondromas are benign, cartilage-forming tumors within the medullary cavity of the bone[1–3]. Individuals with enchondromatosis syndrome, which encompasses seven major subtypes, develop multiple enchondromas. The most common subtypes are non-hereditary Ollier disease (subtype I) and Maffucci syndrome (subtype II), the latter distinguished by spindle cell hemangiomas that occur in addition to the multiple enchondromas[1,3]. Malignant transformation of enchondromas to chondrosarcomas occurs in >30% of these individuals[3,4].

To date, genome-wide screens have not identified a causative gene for Ollier disease or Maffucci syndrome[5–9]. Individuals with these diseases have an increased incidence of gliomas[3,10] and juvenile granulosa cell tumors[3,11–13]. *IDH1* and, more rarely, *IDH2* mutations in gliomas[14–16] and GNAS-activating mutations in juvenile granulosa cell tumors[17] have been reported. Notably, *IDH1* and *IDH2* mutations were recently reported in solitary central and periosteal enchondromas and chondrosarcomas, including in a few tumors from individuals with enchondromatosis[18]. The possibility that *GNAS* mutations are present in enchondromas and chondrosarcomas has not previously been explored.

We therefore assessed whether mutations in *IDH1*, *IDH2* or *GNAS* may cause enchondroma and spindle cell hemangioma formation in Ollier disease and Maffucci syndrome. Sequence analysis of hotspot mutation sites was performed using lesional tissue from 43 individuals with Ollier disease, and this analysis revealed in 33 subjects (78%) the presence of heterozygous mutations in *IDH1* of c.394C>T (encoding an R132C substitution) or c.395G>A (encoding R132H) (NM_005896.2 for both) or in *IDH2* of c.516G>C (encoding R172S) (NM_002168.2) (**Supplementary Fig. 1a–c**). Among the individuals with Maffucci syndrome, 7 of 13 subjects (54%) carried *IDH1* mutations encoding the R132C substitution. Mutations were absent in DNA isolated from the blood, muscle or saliva of the subjects (**Supplementary Fig. 1b**). Mutations in *GNAS* were absent in the tissues examined.

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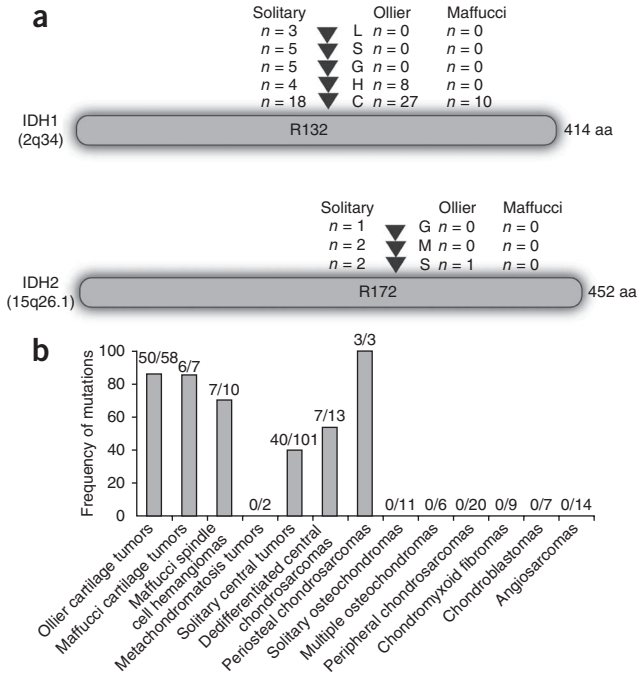


Figure 1 Frequency of IDH1 and IDH2 alterations. (**a**) Distribution of the different Arg132 alterations in IDH1 and Arg172 alterations in IDH2 among the subjects with Ollier disease, Maffucci syndrome and solitary tumors. (**b**) Frequency of somatic heterozygous *IDH* (*IDH1* and *IDH2*) mutations in tumors of subjects with Ollier disease or Maffucci syndrome in comparison to different subtypes of solitary cartilaginous tumors and angiosarcomas.

An additional eight tumors had sub-threshold peaks at the position in *IDH1* expected to encode mutations resulting in R132C or R132H substitutions, suggesting that the mutant allele might be present in a small subpopulation of the tumor cells at the limit of or below the level of detection of Sanger sequencing. We therefore performed a hydrolysis probe assay, which is capable of detecting mutant allele frequencies as low as 1%, to look for *IDH1* mutations encoding R132C or R132H[19,20]. Mutations were confirmed in seven of eight tumors (**Supplementary Fig. 1d–g**), and there was insufficient DNA from the eighth tumor for analysis. Thus, in total, 35 of 43 (81%) and 10 of 13 (77%) subjects with Ollier disease and Maffucci syndrome, respectively, had *IDH1* or *IDH2* mutations (**Fig. 1a**, **Table 1** and **Supplementary Table 1**). The frequency of mutations in tumors is shown in **Figure 1b**.

Other subtypes of enchondromatosis syndromes are known to be caused by mutations in *PTPN11* (metachondromatosis)[21,22] and *ACP5* (spondyloenchondrodysplasia)[23,24] and by *PTHLH* duplication (symmetrical enchondromatosis)[25]. Mutations in *PTH1R*, which encodes a protein involved in enchondral bone formation, are found in ~8% of individuals with Ollier disease but not in those with Maffucci syndrome[5–7]. Previously, an absence of *PTPN11* mutations was shown in the current cohort of individuals[22]. In the current study, we did not detect *PTH1R* mutations in a screen of 35 subjects with Ollier disease or Maffucci syndrome. Analysis performed using a custom-made Agilent tiling array (**Supplementary Table 2**) did not show evidence of loss or gain of *IDH1*, *IDH2*, *PTHLH*, *PTPN11*, *PTH1R*, *EXT1*, *EXT2* or *ACP5*. Thus, even though individuals with enchondromatosis syndromes have overlapping clinical features, they seem to be genetically discrete entities, with the exception of Ollier disease and Maffucci syndrome, which we have now shown to both contain *IDH1* or *IDH2* mutations.

As Ollier disease and Maffucci syndrome are not inherited and enchondromas are often unilateral, we hypothesized that mutations may occur in a somatic mosaic fashion. Fourteen of 16 subjects (88%) possessed identical mutations, including rare variants, in more than one tumor (**Supplementary Table 1**). We additionally

Table 1 Results of *IDH1* and *IDH2* mutation analysis

	Total	Gender (M:F) (median age in years)	Number with *IDH1* mutations (%)	Number with IDH1 R132C (*IDH1* CGT>TGT) (%)	Number with R132H (*IDH1* CGT>CAT) (%)	Number with *IDH2* mutation (%)	Total with *IDH1* or *IDH2* mutation (%)
Ollier disease							
Number of subjects	43	21:21[a] (24)	34 (79)			1 (2)	35 (81)
Enchondroma	25		22 (88)	15 (68)	7 (32)	0	22 (88)
Chondrosarcoma grade I	23		20 (87)	18 (90)	2 (10)	0	20 (87)
Chondrosarcoma grade II	8		5 (63)	5 (100)	0	1 (12)	6 (75)
Chondrosarcoma grade III	2		1 (50)	1 (100)	0	1 (50)	2 (100)
Total number of tumors	58		48 (83)	39 (81)	9 (19)	2 (3)	50 (86)
Maffucci syndrome							
Number of subjects	13	5:8 (15)	10 (77)			0	
Enchondroma	5		4 (80)	4 (100)	0	0	
Chondrosarcoma grade I	1		1 (100)	1 (100)	0	0	
Chondrosarcoma grade II	1		1 (100)	1 (100)	0	0	
Spindle cell hemangioma	10		7 (70)	7 (100)	0	0	
Total number of tumors	17		13 (76)	13 (100)	0	0	
Solitary tumors							
Enchondroma	9		3 (33)	2 (67)	1 (33)	2 (22)	5 (56)
Central chondrosarcoma grade I	20		7[b] (35)	2 (29)	2 (29)	0	7 (35)
Central chondrosarcoma grade II	57		18[b] (32)	9 (50)	1 (6)	3 (5)	21 (37)
Central chondrosarcoma grade III	15		7[b] (47)	5 (71)	0	0	7 (47)
Dedifferentiated chondrosarcoma	13		6[b] (46)	3 (50)	1 (17)	1 (8)	7 (54)
Periosteal chondrosarcoma	3		3 (100)	3 (100)	0	0	3 (100)

[a]Unknown gender for one subject. [b]Other types of mutations present beyond those encoding R132C or R132H.

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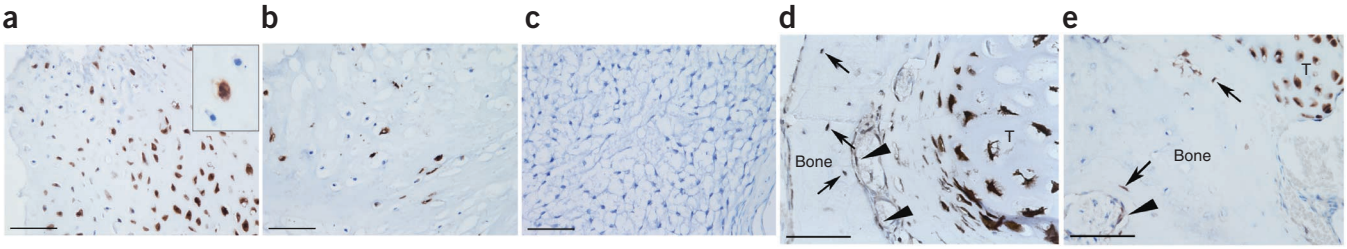


Figure 2 Immunostaining for mutant IDH1 R132H protein. (**a**,**b**) Enchondroma (L1490) of subject with Ollier disease showing strong cytoplasmic and nuclear staining for IDH1 R132H. Note the mixture of cells expressing wild-type and mutant IDH1 indicating intraneoplastic mosaicism. Overall, the percentage of tumor cells positive for mutant IDH1 ranged from 50–95%. Inset shows viability of the negatively stained cells at higher magnification. (**c**) Grade II chondrosarcoma is negative for IDH1 R132H expression. (**d**,**e**) Enchondromas from subjects with Ollier disease showing occasional cells positive for mutant IDH1 in the surrounding normal bone. Some positively stained osteocytes (arrows) and osteoblasts (arrowheads) are seen. T, tumor tissue (scale bars, 5 μm).

used immunohistochemistry to determine the distribution of the IDH1 R132H mutant protein. Of 68 tumors from subjects with Ollier disease, 17 (25%) showed mutant protein expression, whereas 51 (75%) were negative (**Fig. 2** and **Table 2**). Within tumors that were positive for IDH1 R132H staining, we observed a mixture of cells that did and did not express the mutant protein (cells were of the same histologic type and therefore did not include entrapped or supporting elements), a pattern we refer to as intraneoplastic mosaicism (**Fig. 2a**,**b**). Within these tumors, the percentage of tumor cells staining positive for IDH1 R132H ranged from 50% to 95%. Intraneoplastic mosaicism has also been described for other benign bone tumors. In fibrous dysplasia, experimental evidence showed that both normal cells and those with mutations in *GNAS* were needed to develop fibrous dysplasia–like lesions[26]. Also, in osteochondromas, which are benign cartilaginous tumors arising at the surface of the bone that are caused by mutations in *EXT1* or *EXT2*, a mixture of cells with wild-type *EXT* and cells with mutations in *EXT* was observed[27–30]. The EXT protein is involved in heparan sulfate biosynthesis, and it is hypothesized that cells with mutations in *EXT* that are therefore deficient in heparan sulfate need heparan sulfate from neighboring cells for cellular signaling and survival[31,32].

We additionally studied the surrounding normal tissue of Ollier disease–derived and solitary tumors expressing mutant IDH1 R132H protein and observed a very low frequency (on average <1%) of mutant protein in osteoblasts, osteocytes, adipocytes and fibroblasts (**Fig. 2d**,**e**). We were able to perform the hydrolysis probe assay on DNA isolated from one normal bone of a subject with Ollier disease, and we did not detect any changes at the *IDH1* locus. Mutant IDH1 R132H protein was absent in 12 bones resected for reasons other than chondrosarcoma removal as well as in normal growth plates and articular cartilage (**Table 2**). Therefore, our current data support a model of somatic mosaicism, similar to that described for polyostotic fibrous dysplasia in which somatic mosaic mutations in *GNAS* have a causative role[33,34]. Unfortunately, the nature of the samples (decalcified, paraffin-embedded bone) and the occurrence of mutations in single, scattered cells did not allow verification of this theory using other techniques. However, the antibody recognizing IDH1 R132H was shown to be highly reliable for glioma diagnosis[35] and correlated well with sequence analysis in our series.

Twelve tumors were negative for *IDH1* or *IDH2* hotspot mutations. For 5 of these, all exons were sequenced and no mutations were identified. This finding was not surprising, as only *IDH1* mutations affecting Arg132 and *IDH2* mutations affecting Arg140 or Arg172 have been identified in other *IDH*-associated tumors. It is possible that, because of intralesional mosaicism, only a small fraction of tumor cells contains the *IDH1* or *IDH2* hotspot mutations, which may be below the detection level of the techniques used to identify them. Alternatively, mutations in other genes such as *TET2*, in which mutations are mutually exclusive to those in *IDH1* or *IDH2* in acute myeloid leukemia (AML)[36], might be involved[18,37].

Recently, point mutations in *IDH1* or *IDH2* were reported in 56% of solitary central and periosteal cartilaginous tumors[18], and the data within our control group are in concordance with these findings. In total, 40 of 101 (40%) solitary central tumors, 7 of 13 (54%) dedifferentiated chondrosarcomas and 3 of 3 (100%) periosteal chondrosarcomas had *IDH1* or *IDH2* mutations (**Fig. 1b** and **Table 1**). In six additional

Table 2 Immunohistochemical detection of mutant IDH1 R132H protein

	Total number of tumors	IDH1 R132H–positive tumors/total tumors analyzed (%)
Ollier disease		
Enchondroma	46	14/43[a] (32)
Chondrosarcoma grade I	22	3/17[a] (18)
Chondrosarcoma grade II	10	0/8[a]
Maffucci syndrome		
Enchondroma	9	0/9
Spindle cell hemangioma	14	0/14
Solitary tumors		
Enchondroma	19	4/19 (21)
Central chondrosarcoma grade I	42	4/38[a] (10)
Central chondrosarcoma grade II	36	1/32[a] (3)
Central chondrosarcoma grade III	14	0/11[a]
Central dedifferentiated chondrosarcoma	26	1/24[a] (4)
Periosteal chondrosarcoma	6	1/6 (17)
Solitary osteochondroma	20	0/17[a]
Multiple osteochondroma	7	0/7
Peripheral chondrosarcoma	45	0/35[a]
Peripheral dedifferentiated chondrosarcoma	16	0/16
Conventional hemangioma	3	0/3
Hemangioendothelioma	2	0/2
High grade angiosarcoma of bone	44	0/44
High grade angiosarcoma of soft tissue	22	0/22
Controls		
Normal growth plate	3	0/3
Articular cartilage	3	0/3
Normal bone	12	0/12

[a]Not all tumors included were evaluated due to loss of tissue on the tissue microarrays.

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Table 3 IDH1 or IDH2 alterations in solitary central chondrosarcoma cell lines and primary tissue culture

Cell line	Tumor type	Tumor grade	Passage	IDH1 alteration	IDH2 alteration	Reference
SW1353	Solitary central	CSII	12	WT	R172S	ATCC
JJ012	Solitary central	CSII	15	R132G	WT	51
CH2879	Solitary central	CSIII	16	G105G	WT	52
OUMS27	Solitary central	CSIII	18	WT	WT	53
L835	Solitary central	CSIII	38	R132C	WT	Established by the authors
C3842	Ollier disease	CSII	32	WT	WT	54
L2975	Dedifferentiated CS		31	WT	R172W[a]	Established by the authors
NDCS1	Dedifferentiated CS		12	WT	WT	55

CS, chondrosarcoma; WT, wild type.

[a]L2975 had a homozygous mutation in *IDH2* encoding R172W.

tumors, the mutant allele seemed to be present below the detection level of Sanger sequencing. *IDH1* or *IDH2* mutations were absent in other subtypes of cartilaginous tumors, in angiosarcomas (**Fig. 1b**) and in DNA isolated from subjects' blood. Immunostaining for IDH1 R132H protein on tissue microarrays (TMAs) containing cartilaginous and vascular tumor samples confirmed that the expression of mutant IDH1 was restricted to central, dedifferentiated and periosteal cartilage tumors, whereas all other tumors lacked mutant expression (**Table 2**). Of note, four of eight solitary chondrosarcoma cell lines carried different types of mutations in *IDH1* or *IDH2* (**Table 3**). To the best of our knowledge, no representative cell lines with *IDH1* or *IDH2* mutations were previously available. *IDH1* or *IDH2* mutations were more frequently found in solitary central tumors located in the hands and feet (11 of 14 tumors) compared to those located in long and flat bones (28 of 84 tumors) ($P = 0.006$, Pearson's χ^2 test), which was also reported previously[18]. This correlation was absent in Ollier disease (20 of 22 tumors from the hands and feet compared to 28 of 34 tumors from long or flat bones, $P = 0.5$, Pearson's χ^2 test). Whereas in gliomas, mutations in *IDH1* or *IDH2* predict a favorable outcome[38], we found no significant prognostic value of these mutations in solitary central cartilaginous tumors using multivariate analysis (Cox regression, P value = 0.3).

IDH1 and *IDH2* mutations have also been reported at lower frequencies in various other cancers, such as AML (8%)[39,40], prostate cancer (2.7%)[40,41], paragangliomas (0.7%)[40,42] and thyroid carcinoma (16%)[43]. The high mutation frequency in enchondromas and the fact that these mutations occur early suggest a causal rather than a bystander role for *IDH1* and *IDH2* mutations in tumorigenesis in Ollier disease and Maffucci syndrome. In gliomas, mutations in *IDH1* and *IDH2* lead to a gain of function, causing the production of 2-hydroxyglutarate (2HG), a structural analog of α-ketoglutarate (α-KG), which thereby reduces α-KG production[44]. In AML, it was demonstrated that mutant IDH expression results in DNA hypermethylation and impairment of hematopoietic differentiation[36], and in gliomas, the presence of an *IDH1* mutation was strongly associated with hypermethylation[45]. Therefore, we used Illumina HumanMethylation27 BeadChips to examine possible differences in methylation between enchondromas with ($N = 8$) and without ($N = 4$) *IDH1* mutations, as determined by Sanger sequencing. Unsupervised clustering of the 2,000 most variable CpG methylation sites gave two subgroups (**Fig. 3**). One of these subgroups showed an overall higher methylation at the examined CpG sites, a phenotype that is similar to the CpG island methylator phenotype (CIMP) described in colon carcinoma and glioblastoma[45,46]. All but one enchondromas with an *IDH1* mutation were positive for this CIMP. Supervised clustering analysis indicated that 797 CpG sites were differentially methylated by more than 20% (with $P < 0.05$) between enchondromas with and without *IDH1* mutations. Of note, 710 of these differentially methylated CpG sites (89.1%) were methylated in the enchondromas with *IDH1* mutations (**Supplementary Table 3**). These results are in line with the hypothesis that *IDH1* mutations induce methylation and thus contribute to CIMP occurence[36].

To assess the effect of *IDH1* or *IDH2* mutation on mRNA expression levels in cartilaginous tumors, we performed whole-genome gene expression analysis using Illumina Human-6 v3 arrays. High-quality mRNA was available for only three tumors in which no mutation was found ($N = 1$) or in which mutations occurred at a frequency below the threshold of detection with Sanger sequencing (thus possibly carrying a low percentage of cells with mutations) ($N = 2$). Comparison of mRNA expression in these tumors with that from 18 tumors with clearly detectable *IDH1* or *IDH2* mutation using LIMMA analysis revealed 36 differentially expressed probes encoding 33 genes (**Supplementary Table 4**). Of these 33 genes, 32 were downregulated in the tumor samples with an *IDH1* or *IDH2* mutation. There was no overlap between the affected genes identified by methylation or expression analysis.

One of the most differentially methylated genes was *DLX5*, for which there was a trend of downregulation in the expression data comparing enchondromas from subjects with Ollier disease and controls. However, this difference was not significant (adjusted P value = 0.3, **Supplementary Fig. 2**). The controls consisted of two growth plates and four articular or rib cartilage samples. *DLX5* encodes a homeodomain

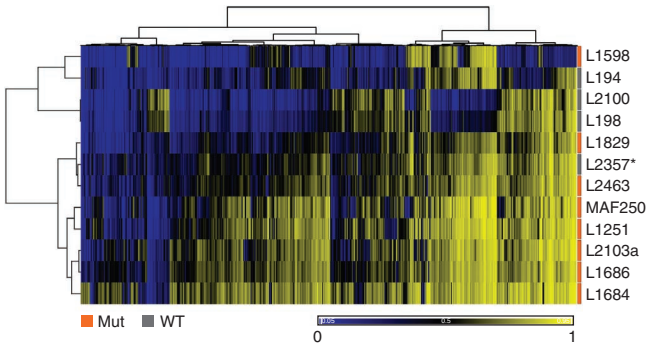


Figure 3 CpG island methylator phenotype in enchondromas with *IDH1* mutations. Heatmap depicting unsupervised clustering analysis of the 2,000 most variable CpG sites of enchondromas with (orange, $N = 8$) and without (gray, $N = 4$) *IDH1* mutation. The level of DNA methylation (beta value) for each probe (columns) in each sample (rows) is represented by color, ranging from 0 (0% methylation, blue) to 1 (100% methylation, yellow). The asterisk indicates sample L2357 in which the mutated allele of *IDH1* encoding R132G was detected in a subpopulation of cells. However, the mutation escaped detection by Sanger sequencing, and therefore the sample is labeled wild type.

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transcription factor that is a cell-autonomous positive regulator of chondrocyte maturation during endochondral ossification, promoting the conversion of immature proliferating chondrocytes into hypertrophic chondrocytes[47,48]. The Dlx5 protein also induces expression of Runx2 and osterix, promoting osteogenic differentiation[49,50]. Future studies should reveal whether downregulation of *DLX5* through methylation as a consequence of *IDH1* mutation delays hypertrophic differentiation of chondrocytes and inhibits subsequent osteogenic differentiation, thereby leaving clusters of proliferating chondrocytes behind.

In summary, we report a large multi-institutional series demonstrating somatic heterozygous point mutations in *IDH1* or, rarely, in *IDH2* in tumor tissues of 81% of subjects with Ollier disease and 77% of those with Maffucci syndrome, and we provide evidence for intraneoplastic and somatic mosaicism. Future studies using deep-sequencing approaches should reveal whether the percentage of individuals carrying somatic mosaic mutations in *IDH1* or *IDH2* is even higher than that detected in our series or whether other genes are involved. We show the *IDH1* mutation to be associated with hypermethylation and downregulation of several genes. Future studies will examine whether there is a causal effect, and it will be of great interest to assess how this dysregulation leads to enchondroma and spindle cell hemangioma formation. Finally, this is the first report of four chondrosarcoma cell lines carrying *IDH1* or *IDH2* mutations, thereby providing good *in vitro* models for functional studies to dissect the role of IDH1 and IDH2 in Ollier disease and Maffucci syndrome, as well as allowing studies of their function in tumorigenesis in general.

URLs. EuroBoNeT, http://www.eurobonet.eu/.

METHODS

Methods and any associated references are available in the online version of the paper at http://www.nature.com/naturegenetics/.

Accession numbers. MIAME-compliant data from the tiling, expression and methylation arrays have been deposited in the GEO database (GSE30844). Sequence data for *IDH1* and *IDH2* has been deposited in GenBank (NM_005896.2 and NM_002168.2).

Note: Supplementary information is available on the Nature Genetics website.

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AUTHOR CONTRIBUTIONS

The study was designed, written and reviewed by T.C.P. and J.V.M.G.B. Mutation analysis was designed and performed by T.C.P., M.A.J.H.v.R., J.V.M.G.B., K.S., T.v.W. and R.v.E. Immunohistochemistry was conducted and evaluated by T.C.P., M.A.J.H.v.R. and J.V.M.G.B. T.C.P., S.L.J.V., J.G.v.O. and D.M. contributed tissue microarrays. Expression profiling was designed and performed by A.-M.C.-J., T.C.P., J.V.M.G.B. and J.O. and analyzed by J.O. and M.L.K. Methylation profiling was designed by A.-M.C.-J., J.V.M.G.B. and L.S., performed by P.d.A., and the results analyzed by P.d.A. and P.J.F. K.H.N., S.D., L.S., B.T., B.L.-A., M.S.-J., R.S., N.L., L.-G.K., C.G., M.V., L.M.B. and K.C.K. each contributed frozen or paraffin-embedded tissues for multiple subjects with Ollier disease or Maffucci syndrome and acquired data for these individuals. The manuscript was approved by all authors.

COMPETING FINANCIAL INTERESTS

The authors declare no competing financial interests.

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ONLINE METHODS

Subjects and clinical specimens. Fresh-frozen tumor tissues ($N = 60$) of 44 subjects with multiple cartilage tumors (36 individuals with Ollier disease and 8 with Maffucci syndrome) (**Table 1** and **Supplementary Table 1**) were collected from the EuroBoNeT consortium[8] and the Laboratory of Human Molecular Genetics at the de Duve Institute, Université catholique de Lovain. In addition, paraffin-embedded tumor tissues ($N = 15$) from 12 subjects were obtained from the archives of the Children's Hospital Boston. Samples were handled according to the ethical guidelines of the host institutions. All samples were coded and the ethical guidelines described in the "Code for Proper Secondary Use of Human Tissue in The Netherlands" (Dutch Federation of Medical Scientific Societies) were followed in all procedures. Chondrosarcoma samples were graded as described[56]. DNA derived from normal saliva, blood or muscle was available from 12 individuals with Ollier disease. The ages of the subjects were documented at the time of operation. Demographic and survival data were obtained from patient records at the host institutions. Written informed consent was obtained for all study participants from whom normal DNA was included. For the use of tumor tissue, the Code for Proper Secondary Use of Human Tissue in The Netherlands guidelines were followed.

For comparison with other cartilage tumors, we included DNA from solitary enchondromas ($N = 9$), solitary central chondrosarcomas ($N = 92$), central dedifferentiated chondrosarcomas ($N = 13$), periosteal chondrosarcomas ($N = 3$) and 37 peripheral cartilaginous tumors (solitary osteochondroma ($N = 11$), peripheral chondrosarcomas ($N = 20$) and multiple osteochondromas ($N = 6$)), as well as from chondromyxoid fibromas ($N = 9$), chondroblastomas ($N = 7$) and osteochondroma-like lesions of metachondromatosis ($N = 2$). Matching blood-derived DNA was also available from 24 subjects as controls. Additionally, we included DNA from angiosarcomas ($N = 14$) because individuals with Maffucci syndrome have central cartilage tumors combined with vascular tumors. The angiosarcomas, chondromyxoid fibromas and chondroblastomas were analyzed for *IDH1* mutations only. Thus, in total, we analyzed 261 tumors from 242 subjects.

DNA extraction and mutation analysis. Genomic DNA from frozen tumors containing at least 80% tumor cells, as estimated on haematoxylin and eosin–stained frozen sections, and from blood and saliva was isolated as described previously[8]. DNA from paraffin-embedded tissue was isolated after microdissection as described[8]. For cell lines and primary tissue culture, DNA was isolated from cell pellets using the Wizard Genomic DNA Purification kit (Promega), according to the manufacturer's instructions.

PCR amplification was performed on exon 4 of *IDH1* for all the samples. Exon 4 of *IDH2* was amplified in samples without *IDH1* mutation, and exon 8 of *GNAS* was studied in samples without *IDH1* or *IDH2* mutation. To correlate with possible *PTH1R* mutations, we also amplified exon 4 of *PTH1R* for mutations encoding G121E and A122T substitutions, exon 5 for mutations encoding R150C and exon 9 for mutations encoding R255H using DNA from 35 subjects with Ollier disease or Maffucci syndrome. PCR was carried out in a reaction volume of 25 µl, with 10 ng of DNA, 12.5 µl of iQ SYBR green Supermix (Bio-Rad) and 10 pmol M13-tailed primers (sequences provided in **Supplementary Table 5**). PCR was performed in a CFX 96 Real-Time PCR detection system (Bio-Rad), with an initial denaturation step of 5 min at 95 °C followed by 40 cycles of 10 s at 95 °C, 10 s at 60 °C and 10 s at 72 °C. After a final elongation step of 10 min at 72 °C, a melt curve was obtained to evaluate the quality of the PCR products. PCR products were purified using the Qiagen MinElute 96 UF PCR Purification system and eluted in 25 µl of sterile water. PCR amplicons were sequenced by a commercial entity using standard forward and reverse M13 primers (Macrogen). The sequence trace files were analyzed with Mutation Surveyor DNA Variant Analysis software (version 3.97, SoftGenetics).

To validate the mutations in *IDH1* encoding R132C and R132H, we designed hydrolysis probe assays (probe sequences provided in **Supplementary Table 6**), using the Custom Taqman Assay Design Tool (Applied Biosystems). Assays were performed on 144 samples, including tumors derived from subjects with Ollier disease and Maffucci syndrome, as well as solitary cartilaginous tumors, chondrosarcoma cell lines and blood from subjects with Ollier disease. Assays were also performed on negative controls (healthy donor DNA) and no template controls. qPCR was carried out in a reaction volume of 10 µl as described previously[57] in a CFX38 Real-Time PCR Detection System (Bio-Rad), with an initial denaturation step of 10 min at 95 °C followed by 40 cycles of 10 s at 92 °C and 30 s at 60 °C. The quantification cycle (C_q) was used for quality assessment and samples with $C_q > 35$ for the wild-type allele were considered as DNA negative. The threshold for the mutant alleles of *IDHI* encoding R132C (c.394C>T) or R132H (c.395G>A) was set after subtracting the highest background signal from the negative controls.

There was sufficient DNA left from 5 of 12 tumors without mutation to perform sequence analysis for all exons of *IDH1* and *IDH2*. One tumor with an *IDH1* mutation was also sequenced. PCR was performed as described above for exon 4, and primer sequences are listed in **Supplementary Table 5**.

Tiling resolution targeted oligonucleotide arrays. Custom-designed Agilent tiling oligonucleotide array–comparative genomic hybridization analysis consisting of 15,000 probes with a tiling coverage of genes involved in the different types of enchondromatosis syndromes (*IDH1*, *IDH2*, *ACP5*, *PTH1R*, *PTPN11*, *EXT1*, *EXT2* and *PTHLH*) (**Supplementary Table 2**) was performed to detect possible small intragenic losses and gains in these genes. In total, 16 enchondromas and chondrosarcomas from subjects with Ollier disease or Maffucci syndrome, with ($N = 14$) and without ($N = 2$) *IDH1* or *IDH2* mutations were selected. Labeling and hybridization of genomic DNA from freshly frozen tumors and data processing were performed as described[58].

Immunohistochemistry. To examine the protein expression of the IDH1 R132H mutant, immunohistochemistry was performed as described[8] using antibody recognizing IDH1 R132H from Dianova (1:200 dilution in 5% non-fat milk, citrate antigen retrieval and blocking for 30 min with 5% non-fat milk). We used 403 tumors (**Table 2**) on 19 TMAs, for which details were previously published[8,59–61]. Additional samples from subjects with Ollier disease or Maffucci syndrome were collected through the European Musculo-Skeletal Oncology Society (EMSOS), and clinical details for these individuals are described separately[5]. Glioma tissue with a known *IDH1* mutation was used as a positive control, and primary antibody was omitted as a negative control. Only strong cytoplasmic staining combined with nuclear staining was considered a positive result[35]. To study possible mosaicism in the tumors and in surrounding normal tissues, we selected resection specimens from tumors expressing the mutant IDH1 R132H protein ($N = 7$) and stained multiple tissue blocks from different areas. All except nine tumors from subjects with Ollier disease that were used for mutation analysis were also included in the TMAs, and results were confirmed.

Statistical analysis for clinical correlation. Follow-up data were available from 83 subjects with solitary tumors (range 2–335 months, mean 115.23). To investigate the relation of *IDH1* or *IDH2* mutation with the clinical features of the subjects, multivariate survival analysis (Cox regression) and cross-tabulations (Pearson's χ^2 test) were performed using SPSS version 16.0. Statistical analysis was not performed for subjects with Ollier disease because nearly all subjects with available follow-up data had *IDH1* or *IDH2* mutations. All the *P* values reported are two-sided, and $P < 0.05$ was considered to indicate statistical significance.

DNA methylation profiling. A total of 12 samples, which included 8 enchondromas with *IDH1* mutation (4 Ollier enchondromas, 2 Maffucci enchondromas and 2 solitary enchondromas) and 4 enchondromas (1 Ollier enchondroma and 3 solitary enchondromas) without *IDH1* or *IDH2* mutations, were analyzed. Of the 4 enchondromas without *IDH1* mutation, one had cells with mutated *IDH1* encoding the R132G alteration present in a subpopulation, which was below the threshold of detection by Sanger sequencing. Bisulfite treatment was performed using the EZ DNA Methylation kit (Zymo Research). Bisulfite-converted DNA was then hybridized to Illumina HumanMethylation27 BeadChips by following the manufacturer's instructions. Infinium unsupervised clustering analysis was performed using the Ward's clustering algorithm based on Euclidian distance. The 2,000 most variable CpG sites (excluding those on the X and Y chromosomes) were used in the clustering analysis.

Genome-wide gene expression analysis. A total of 21 tumors, including 6 enchondromas and 10 chondrosarcomas (6 grade I and 4 grade II) from subjects with Ollier disease and Maffucci syndrome, as well as 1 solitary enchondroma, 4 solitary chondrosarcomas, grade II, and 6 controls (2 growth plates and 4 normal cartilage), were used. We determined differential expression between tumors with *IDH1* or *IDH2* mutation ($N = 18$) compared to tumors without

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detectable *IDH1*or *IDH2* mutation ($N = 3$) using Sanger sequencing. Two of these samples showed sub-threshold peaks for mutations in *IDH1* encoding R132G and R132C, suggesting that the mutation was present in a minor subpopulation of tumor cells. Expression analysis using Illumina Human-6 v3.0 Expression BeadChips was performed as described previously[8,62,63]. LIMMA analysis[64] was used to determine differential expression between the groups. Probes with Benjamini and Hochberg false discovery rate–adjusted *P* values <0.05 and a log fold change >0.1 were considered to be significantly differentially expressed.

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